

ComeBack AI

The AI Recovery Coach That Helps You Restart

Because Most People Don't Fail — They Just Stop Trying

Full Product Requirements & Technical Document

Build With AI Hackathon · GDG Lagos · 2025

Version 1.0 · Category: Daily Productivity + AI Agents

Google Cloud

Gemini 2.5

Firebase

React + TypeScript

How To Read This Document

Think of this doc like a recovery plan. It tells everyone on the team — the designer, the coder, the business person — exactly what we are building, why abandoned goals are a universal problem, and how AI can help people restart. You do not need to be a tech expert to understand it. Just read it like a story about human behavior.

This document is split into seven clear parts:

- Part 1 — The Problem: Why people abandon goals and why nobody helps them restart
- Part 2 — Our Solution: What ComeBack AI is and why it is different from every productivity app
- Part 3 — The Market: Who needs this, why they need it, and how big the opportunity is
- Part 4 — The Product: Every feature, every screen, and every user journey explained simply
- Part 5 — The Tech: The exact tools, AI agents, and Google services we will use to build it
- Part 6 — Winning Strategy: How we will win the hackathon with our demo and pitch
- Part 7 — Impact & Vision: How ComeBack AI makes humans better and changes lives

For Team Members: Non-technical members should focus on Parts 1-4, 6, and 7. Developers and engineers should read everything, with special attention to Part 5. Everyone should read Part 7 before the pitch — it is our emotional core.

PART 1 — The Problem: Silent Abandonment

1.1 The Story Nobody Talks About

Meet Chidi. He is 23 years old, lives in Lagos, and wants to become a frontend developer. On January 1st, he creates a plan:

- Month 1: Learn HTML and CSS
- Month 2: Master JavaScript
- Month 3: Build React projects
- Month 4: Apply for jobs

He starts strong. For the first week, he studies every day. He completes tutorials. He takes notes. He feels motivated.

Then life happens. His laptop breaks. He gets sick. Work becomes overwhelming. By Week 3, he has missed 5 days. By Week 6, he has stopped completely.

Nobody notices. Nobody checks on him. Nobody helps him restart.

Six months later, Chidi still wants to be a developer. But he feels like a failure. He thinks: "I already tried and quit. Why would this time be different?"

This is Silent Abandonment. People quietly give up on goals. Not because they do not want them anymore. But because they missed a few days, felt like failures, and had no system to help them recover. This happens to millions of people every single day.

1.2 The Real Problem — Broken Down Simply

Most productivity apps assume people fail because they do not know what to do. This is false. Most people already know what to do.

Who	They Know They Should...	But They Don't Because...
Students	Study consistently, complete assignments, learn new skills	They get overwhelmed, lose motivation, and quietly stop trying
Developers	Finish online courses, build projects, contribute to open source	They start strong but abandon courses halfway through

Job Seekers	Apply to jobs daily, update resume, practice interviews	Rejection fatigue makes them stop applying
Entrepreneurs	Follow up with customers, post on social media, track finances	They get busy and let important tasks slide
Fitness Enthusiasts	Exercise regularly, eat healthy, track progress	They miss a few days and feel like they failed

The Pattern: The problem is not planning. The problem is not knowledge. The problem is that people do not have a system to help them recover when they inevitably fall off track. Every productivity app helps you start. Almost none help you restart.

1.3 Why Existing Apps Fail At This

We looked at every major productivity app. Here is what they get wrong:

App	What It Does Well	The Fatal Flaw	What Happens When You Fail
Notion AI	Great for organization and planning	Does not push you or check on you	Your plan sits there unused. Nobody notices.
Todoist	Simple task management	No accountability or recovery system	Tasks pile up. You feel guilty. You stop opening the app.
Motion	AI scheduling	Expensive (\$34/month) and punishes failure	Missed tasks get rescheduled but you feel like you are falling behind.
ChatGPT	Gives great advice	No long-term memory or follow-up	You get advice once. Then you are on your own.
Habitica	Gamification makes it fun	Becomes repetitive and punishes missed days	Your character dies. You feel worse. You quit the app.
Duolingo	Streaks work for language learning	Only works for one specific use case	Break your streak and the app guilt-trips you.

The Gap: All of these apps reward success. None of them are built for failure. But failure is where users spend most of their time. This is the gap ComeBack AI fills.

PART 2 — Our Solution: Meet ComeBack AI

2.1 What Is ComeBack AI?

ComeBack AI is not another task manager. It is not another planner. It is not another productivity app that assumes you will succeed if you just try harder.

ComeBack AI in one sentence: ComeBack AI is an AI-powered recovery coach that helps students, professionals, and learners restart abandoned goals through intelligent recovery plans, compassionate accountability, and personalized coaching that understands failure is part of the journey.

2.2 The Core Insight That Makes Us Different

Most apps reward success. ComeBack AI is built for failure.

People do not fail because they miss one day. They fail because they think missing one day means they failed completely. ComeBack AI breaks this cycle.

2.3 How ComeBack AI Works — The Simple Version

1. You tell ComeBack AI your goal: "I want to become a frontend developer by December 2026"
2. AI creates a realistic plan with monthly milestones and daily targets
3. Every day, AI checks in: "Did you complete today's task?"
4. When you miss days (and you will), AI does not punish you — it creates a Recovery Plan
5. Recovery Plan starts small (15 minutes) and gradually rebuilds your momentum
6. AI tracks your progress, celebrates small wins, and adjusts the plan based on your real behavior

The Magic Moment: When you come back after 12 days of silence, other apps say "You failed." ComeBack AI says "Welcome back. Let's rebuild your momentum together." This changes everything.

2.4 The Three Things That Make ComeBack AI Unique

1. Recovery-First Design	2. Compassionate AI	3. Behavioral Intelligence
Built specifically for people who have abandoned goals. The entire product is designed around the restart moment, not the start moment.	AI that understands failure is normal. No guilt trips. No punishment. Just intelligent support that meets you where you are.	AI learns your patterns — when you struggle, what causes burnout, what helps you recover — and adapts your plan accordingly.

PART 3 — The Market: Who Needs This And Why

3.1 Our Target Users — Who We Are Building For

ComeBack AI is built for anyone who has ever started a goal and quietly stopped. But we are focusing on three specific groups in Nigeria first:

Target Group	Who They Are	Their Biggest Pain	Why They Need ComeBack AI
Tech Learners	18-30 year olds learning frontend, backend, DevOps, data analytics, cybersecurity to get tech jobs	Start learning roadmaps but abandon them after 2-3 weeks. Feel like failures. Do not know how to restart.	AI creates recovery plans that rebuild momentum gradually. No judgment, just intelligent support.
University Students	Undergrads and postgrads trying to balance academics, skills, internships, and personal projects	Get overwhelmed by multiple goals. Abandon some quietly. No system to help them recover.	AI prioritizes goals, detects burnout early, and helps them restart with realistic expectations.
Young Professionals	22-35 year olds pursuing certifications, career growth, side hustles while working full-time	Buy courses but never finish them. Start side projects but abandon them. Feel stuck.	AI tracks their real behavior, adjusts timelines, and creates recovery plans that fit their actual life.

3.2 Why We Are Targeting These Groups

Reason 1: Massive Market Size

- Nigeria has over 200 universities with 2+ million students
- Over 500,000 Nigerians are actively learning tech skills (frontend, backend, data, etc.)
- Millions of young professionals are pursuing career growth and side hustles
- This is a market of 5-10 million people in Nigeria alone

Reason 2: High Pain Point

- 92% of people abandon New Year resolutions by February
- 70% of online course buyers never finish the course
- Most tech learners abandon their roadmaps within 30 days
- The pain of silent abandonment is universal and deeply felt

Reason 3: No Existing Solution

- No app is specifically built for goal recovery
- All productivity apps assume you will succeed if you plan better
- Nobody is solving the restart problem — this is a blue ocean

Reason 4: Viral Potential

- Everyone has abandoned a goal — this is universally relatable
- People will share their comeback stories on social media
- Word-of-mouth will be strong because the problem is emotional

3.3 Market Research — The Numbers That Prove This Is Real

Market Fact	Why It Matters for ComeBack AI
92% of people abandon New Year resolutions by February	Goal abandonment is not a personal failure — it is a universal human behavior that needs a systemic solution.
70% of online course buyers never finish the course	People are spending money on goals they abandon. They need help finishing, not just starting.
Global productivity app market = \$96.36 billion by 2030	Huge market but nobody is solving the recovery problem. We can carve out a significant niche.
Nigeria has 2+ million university students	Our primary target market is large, growing, and digitally connected.
500,000+ Nigerians learning tech skills actively	Tech learners are our ideal early adopters — they understand AI and need accountability.
Average person has 3-5 abandoned goals at any time	This is not a one-time problem. People will use ComeBack AI repeatedly for different goals.

PART 4 — Unique Selling Points & Human Impact

4.1 What Makes ComeBack AI Truly Unique

USP #1: Recovery-First, Not Planning-First Every productivity app helps you start. ComeBack AI is the only app built specifically for the restart moment. The entire product is designed around helping you recover from abandonment, not preventing it.

USP #2: Compassionate AI, Not Punishing AI Other apps break your streak, guilt-trip you, or make you feel like a failure. ComeBack AI says "Welcome back" and creates an intelligent recovery plan. No judgment. Just support.

USP #3: Behavioral Intelligence ComeBack AI learns your patterns — when you struggle, what causes burnout, what time of day you are most productive — and adapts your plan in real-time. It is not a static planner. It is a dynamic coach.

USP #4: Voice-First Interaction You can talk to ComeBack AI like a real coach. "I have been tired all week." AI responds with empathy and adjusts your plan. No typing. No forms. Just conversation.

4.2 How ComeBack AI Makes Humans Better

This is not just about productivity. This is about changing how people relate to failure and growth. Here is the deeper human impact:

Impact 1: Reframes Failure As Part Of The Journey

Most people think: "I missed 5 days. I failed. I should give up."

ComeBack AI teaches: "You missed 5 days. That is normal. Let us restart with a plan that works for where you are right now."

This mental shift is life-changing. People stop seeing themselves as failures and start seeing themselves as humans who are learning and growing.

Impact 2: Builds Self-Compassion

The AI is compassionate by design. It never guilt-trips. It never punishes. It always meets you where you are.

Over time, users internalize this compassion. They become kinder to themselves. They stop beating themselves up for being human.

Impact 3: Creates Sustainable Habits

Most apps push you to do more. ComeBack AI helps you do what is sustainable.

When you are burned out, AI reduces your targets. When you are energized, AI increases them. This creates habits that last because they are built on your real capacity, not an idealized version of yourself.

Impact 4: Reduces Shame And Isolation

Silent abandonment is lonely. People feel like they are the only ones who quit.

ComeBack AI normalizes failure. It shows you that everyone struggles. You are not alone. This reduces shame and makes it easier to ask for help.

Impact 5: Increases Goal Completion Rates

The ultimate measure: Do people actually finish their goals?

Early research shows that people with recovery systems are 3-5x more likely to complete goals than people without them. ComeBack AI is that recovery system.

The Long-Term Vision: In 5 years, we want people to say: "I used to abandon goals all the time. Then I found ComeBack AI. Now I finish what I start — not because I am perfect, but because I have a system that helps me restart when I fall."

PART 5 — The Product: Features & User Experience

5.1 Core Features — What We Are Building

Feature 1: Goal Creation & AI Planning

User tells ComeBack AI their goal in plain language:

- "I want to become a frontend developer by December 2026"
- "I want to lose 10kg in 6 months"
- "I want to finish my AWS certification by March"

AI generates:

- Monthly milestones (Month 1: HTML, Month 2: CSS, Month 3: JavaScript)
- Weekly targets (This week: Complete 3 HTML tutorials)
- Daily tasks (Today: Watch 1 tutorial video, build 1 simple page)

Feature 2: Daily Check-Ins

Every day, AI asks: "Did you complete today's task?"

- Yes → AI celebrates and prepares tomorrow's task
- Partially → AI adjusts tomorrow's task to be more realistic
- No → AI asks why and offers support

Check-ins can be done via:

- Text input
- Voice message (powered by Gemini)
- Quick tap buttons (Yes/No/Partially)

Feature 3: The Recovery Engine (THE CORE FEATURE)

This is what makes ComeBack AI unique. When you miss days, AI creates a Recovery Plan:

Example: You missed 12 days.

Other apps: "You broke your streak. Start over."

ComeBack AI: "You missed 12 days. That is okay. Here is your Recovery Plan:"

- Day 1: 15 minutes (just show up)
- Day 2: 30 minutes (build momentum)
- Day 3: 45 minutes (getting back to normal)
- Day 4: Back to your regular schedule

The AI gradually rebuilds your momentum instead of expecting you to jump back to 100%.

Feature 4: Burnout Detection

AI monitors your behavior patterns:

- Declining task completion rate
- Increasing "No" responses
- Longer gaps between check-ins

When AI detects burnout, it proactively suggests:

- Lighter targets for the next week
- A rest day
- Breaking big tasks into smaller ones

Feature 5: Progress Journal

AI generates weekly summaries:

- "This week you completed 4 out of 7 tasks. That is 57% completion — up from 40% last week. You are improving!"
- Shows trends over time
- Celebrates small wins
- Identifies patterns (e.g., "You struggle on Mondays but do great on Wednesdays")

Feature 6: Voice Coaching

User can talk to AI like a real coach:

- User: "I have been really tired this week"
- AI: "I hear you. Let us reduce today's target to just 15 minutes. Sometimes showing up is enough."

Powered by Gemini 2.5 with empathetic system prompts.

PART 6 — The Tech: How We Are Building It

6.1 Technology Stack (Google-First)

Component	Tool	Alternative Rejected	Why We Chose This
Frontend	React + TypeScript + MUI	Vue, Angular, plain HTML	React is fast to build with. TypeScript prevents bugs. MUI gives us beautiful components out of the box.
AI Engine	Google Gemini 2.5 via AI Studio	OpenAI GPT-4, Claude	Hackathon requirement. Gemini 2.5 is fast, has great function calling, and free tier is generous.
Database	Firebase Firestore	PostgreSQL, MongoDB	Real-time updates for check-ins. Serverless. Easy to set up. Lives on Google Cloud.
Authentication	Firebase Auth	Auth0, custom JWT	Google Sign-In for quick onboarding. Phone OTP for Nigerian users. Set up in 20 minutes.
Backend	Cloud Functions (Node.js)	Express on Cloud Run	Serverless. Integrates with Firestore. Perfect for hackathon speed.
Hosting	Firebase Hosting	Vercel, Netlify	One-command deploy. CDN included. Free tier is generous.

6.2 The Three AI Agents

Agent 1: Goal Planner Agent

Purpose: Convert user goals into actionable plans.

- Input: "I want to become a frontend developer by December 2026"
- Output: Monthly milestones, weekly targets, daily tasks
- Uses Gemini function calling to structure the plan

Agent 2: Recovery Agent (THE CORE)

Purpose: Create intelligent recovery plans when users miss days.

- Monitors: Days missed, previous completion rate, user's stated reason for missing
- Creates: Gradual recovery plan (15min → 30min → 45min → normal)
- Adjusts: Based on how user responds to recovery plan

Agent 3: Motivation Agent

Purpose: Provide empathetic coaching and detect burnout.

- Analyzes: User's language, completion patterns, check-in frequency
- Detects: Burnout signals (declining completion, negative language)
- Responds: With compassionate messages and lighter targets

6.3 Database Schema

Collection	Fields
users	uid, name, email, createdAt, timezone, preferences
goals	goalId, userId, title, deadline, status (active/paused/completed/abandoned), createdAt, completedAt
milestones	milestoneId, goalId, title, targetDate, status, order
tasks	taskId, milestoneId, title, dueDate, status (pending/completed/skipped), completedAt
checkins	checkinId, userId, goalId, date, status (yes/no/partial), note, createdAt
recovery_plans	planId, goalId, daysMissed, startDate, dailyTargets[], status (active/completed)
progress_journal	journalId, userId, goalId, weekStart, weekEnd, summary, completionRate, aiInsights

6.4 24-Hour Build Plan

Hours	Phase	What Gets Built
0-3h	Setup	Scaffold React app. Set up Firebase (Auth, Firestore, Functions). Connect Gemini API. Deploy skeleton. Everyone runs locally.
3-8h	Core AI	Build Goal Planner Agent. Create goal input screen. AI generates plan. Show milestones and tasks. THIS IS THE HERO FEATURE.

8-13h	Check-Ins	Daily check-in screen. Yes/No/Partial buttons. Voice input (Gemini speech-to-text). Store check-ins in Firestore.
13-18h	Recovery Engine	Recovery Agent: detects missed days, creates recovery plan. Recovery screen shows gradual targets. THIS IS OUR UNIQUE FEATURE.
18-21h	Progress & Polish	Progress Journal (weekly summaries). Burnout detection. Motivation Agent responses. Polish UI.
21-24h	Demo Prep	Seed realistic demo data. Test full user journey. Record backup video. Prepare pitch deck.

PART 7 — Winning Strategy & Demo

7.1 What Judges Will Score On

Criterion	What They Want	How We Win
AI Integration	Is AI core or decoration?	3 distinct agents. AI is the product, not a feature.
Real Problem	Do people actually face this?	Everyone has abandoned a goal. Universally relatable.
Technical Execution	Does it work?	Full Google stack. Live demo with real AI responses.
Demo Quality	Can you show it in 3 min?	Clear story: Goal → Miss days → Recovery Plan → Success.
Market Potential	Could this be a real company?	5-10M users in Nigeria. Global potential. Clear monetization.

7.2 The Demo Script (2 Minutes)

Opening (30 seconds)

*"How many people in this room have started a goal and quietly stopped? Raise your hand.
[Everyone raises hand] Exactly. We all do it. But here is the problem: when you miss a few days, every app makes you feel like a failure. Your streak breaks. You feel guilty. You quit completely. We built ComeBack AI to change that."*

Demo (90 seconds)

- 7. Open ComeBack AI. Show clean home screen.
- 8. Create goal: "I want to become a frontend developer by December 2026"
- 9. AI generates plan in 3 seconds: Month 1 HTML, Month 2 CSS, Month 3 JavaScript
- 10. Show daily check-in: "Did you complete today's task?" → Tap "Yes"
- 11. Fast-forward: User misses 12 days
- 12. Show Recovery Engine: "You missed 12 days. That's okay. Here's your Recovery Plan: Day 1: 15 min, Day 2: 30 min, Day 3: 45 min"
- 13. Show Progress Journal: "This week you improved by 25%"

Closing (30 seconds)

"Most apps reward success. ComeBack AI is built for failure. Because failure is not the end — it is just a pause. We help 5-10 million Nigerians restart abandoned goals. The market is global. The problem is universal. And we are just getting started."

PART 8 — Summary: Why ComeBack AI Matters

8.1 The Core Truth

People do not fail because they are lazy. They fail because they think missing one day means they failed completely. ComeBack AI breaks this cycle.

8.2 What Makes Us Unforgettable

- We are the only app built specifically for goal recovery
- We treat failure as normal, not shameful
- We use AI to create compassionate, intelligent recovery plans
- We solve a problem everyone has experienced
- We make humans better by teaching self-compassion

ComeBack AI — The AI Recovery Coach

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Helping people restart what they never should have stopped.